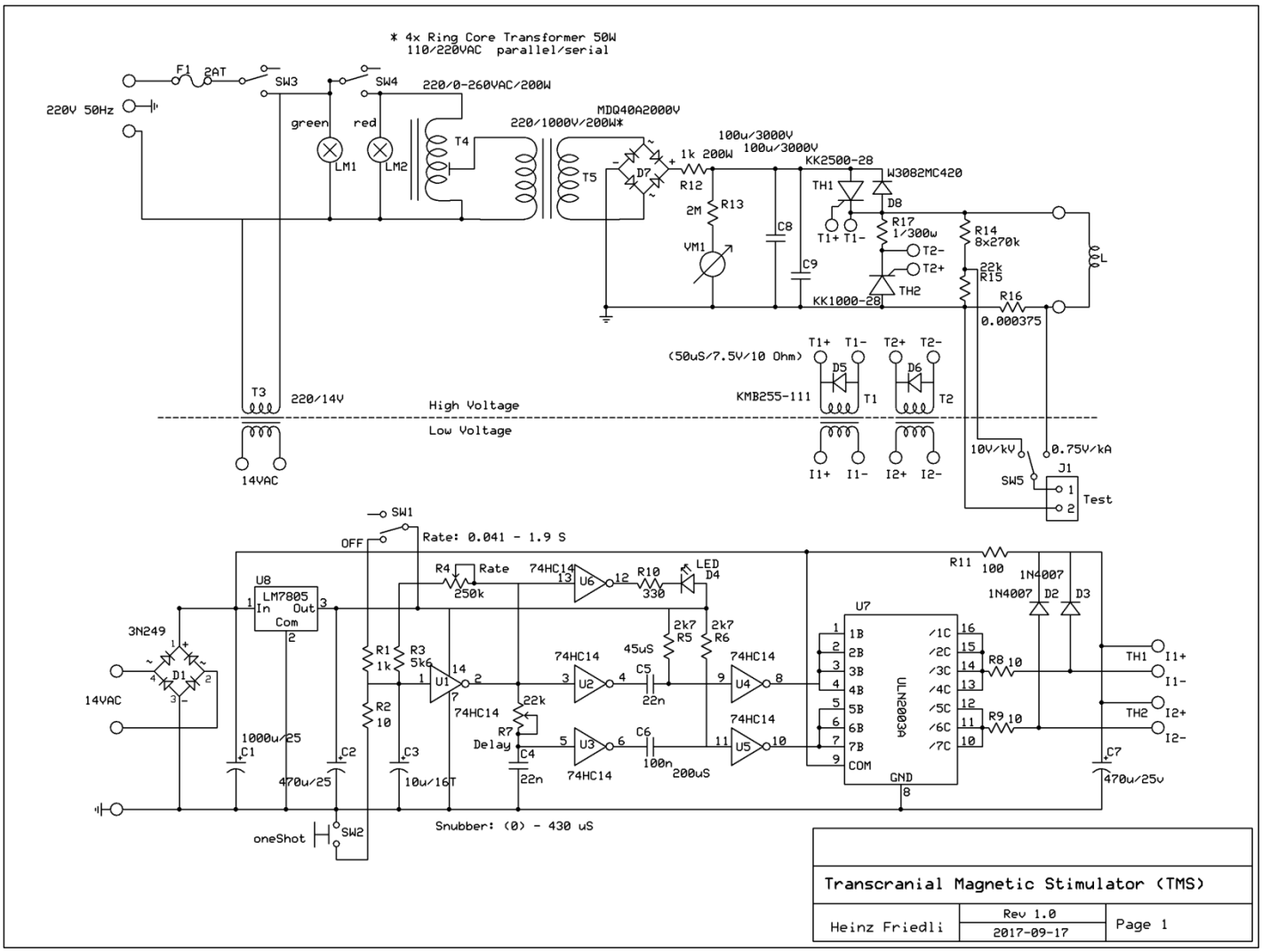
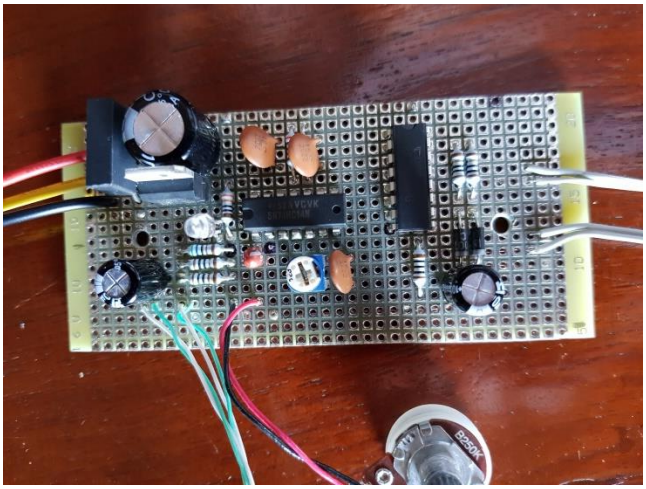
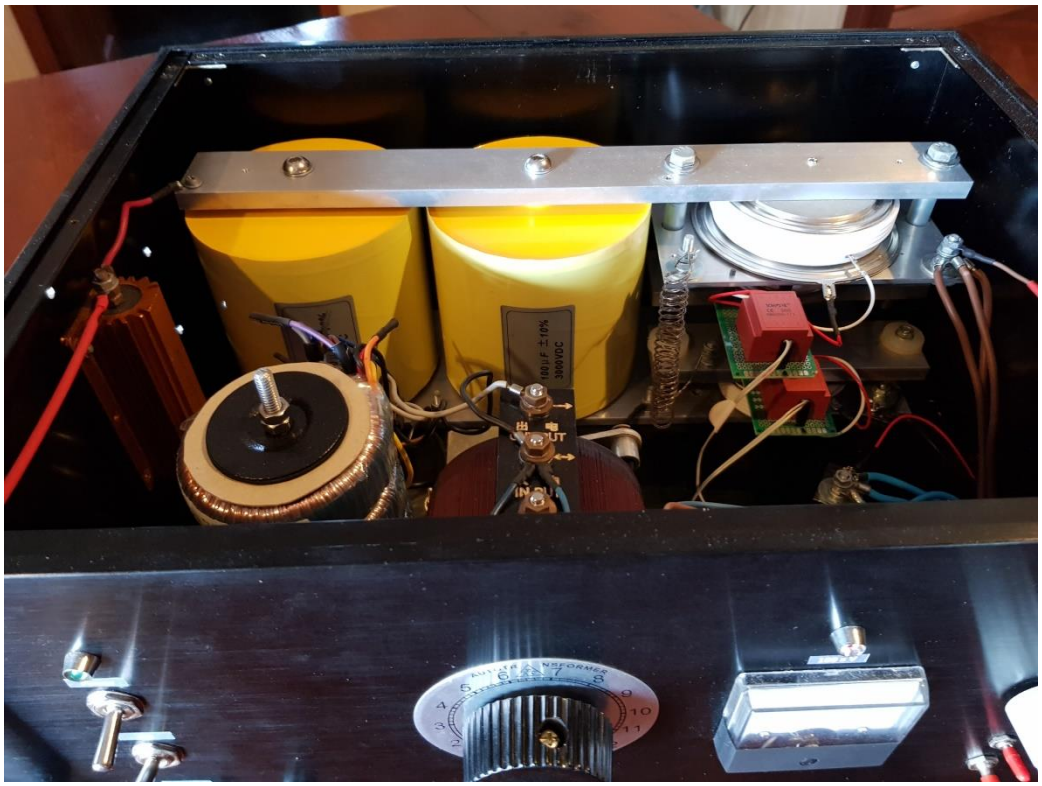
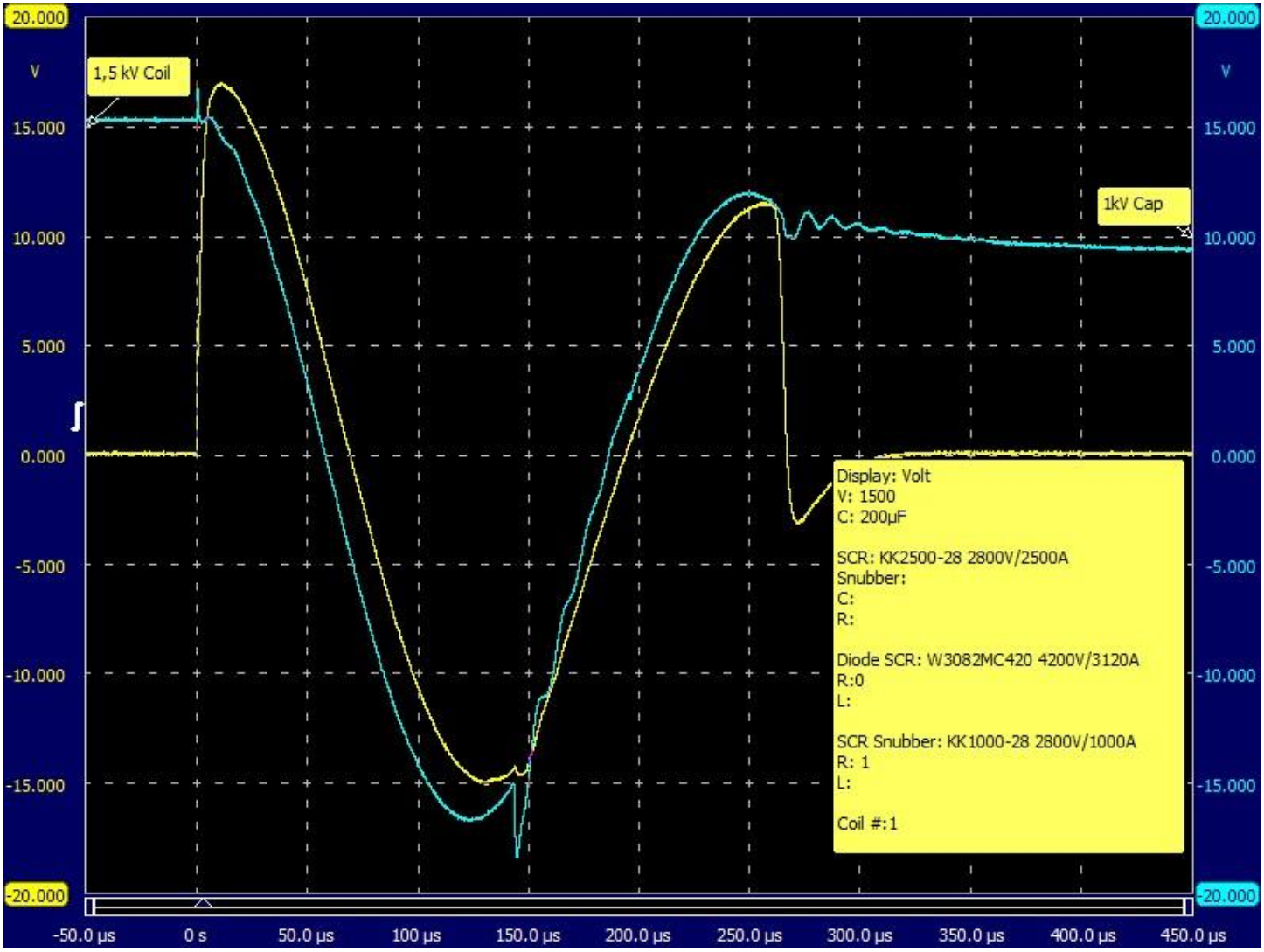
Transcranial Magnetic Stimulator

Design Target: Biphasic Pulse, Action Coil on Ground, simple electronics








20.000
V
15.000
10.000
5.000
0.000
-5.000
-10.000
-15.000
-20.000
1,5 kV Coil
1kV Cap
Display: Volt
V: 1500
C: 200µF
SCR: KK2500-28 2800V/2500A
Snubber:
C:
R:
Diode SCR: W3082MC420 4200V/3120A
R:0
L:
SCR Snubber: KK1000-28 2800V/1000A
R: 1
L:
Coil #:1
-50.0 µs
0 s
50.0 µs
100 µs
150.0 µs
200.0 µs
250.0 µs
300.0 µs
350.0 µs
400.0 µs
450.0 µs

Coil Design:

Coil	#	Wdg	AWG	µH	mΩ [4]	µS/200µF	Volt	kA	kW	**kAt**	Field [2]
2x9 Wdg	1	18	9	9.03	10.00	193.90	1,500	9.00	13,500	**162**	2.30
2x5 Wdg	2	10	9	3.43	5.40	125.00	1,500	15.33	22,995	**153**	2.95
2x9 Wdg	3	18	9	4.15	6.20	140.20	1,500	12.70	19,050	**229**	1.70
2x10 Wdg	4	20	9	13.11	15.00	230.50	1,500	8.80	13,200	**176**	1.80
2x11 Wdg	5	22	10	12.44	14.30	253.00	1,500	7.60	11,400	**167**	1.90
2x9 Wdg	6	18	10 [3]	12.2	14.50	219.30	1,500	8.80	13,200	**158**	2.00

[2] relative

[3] HF Litz Wire 0,1x600

[4] + Cable 125cm (3x 2.5mm2) 5.8mΩ

This was the first actively used coil, AWG10, 2x9 Turns.
After a while sparking flashovers occurred.
The coil was glued with high temperature silicon and wrapped with self-vulcanizing rubber.